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KERTAS PENERANGAN (INFORMATION SHEET)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-4: 2015 CORE ABILITIES	
TAHAP / LEVEL	4(FOUR)	
NO. DAN TAJUK KEBOLEHAN TERAS <i>ICORE ABILITY NO. AND TITLE</i>	CA08 SYSTEM TECHNOLOGY APPLICATION	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	08.01 ADOPT APPROPRIATE SOFTWARE APPLICATION 08.02 COMPLY TO SYSTEM ANALYSIS APPLICATION 08.03 UPDATE IT APPLICATION 08.04 ADHERE TO IT RULES & REGULATIONS	
NO. KOD ICODE NO.	Z-009-4:2015-M08/P(2/4)	Muka Surat / Page : 1 Drpd / of : 11

TAJUK/TITLE: SYSTEM ANALYSIS COMPLIANCE

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on System Analysis Compliance requirements to maintain and ensure information technology applications in the organisation are current.

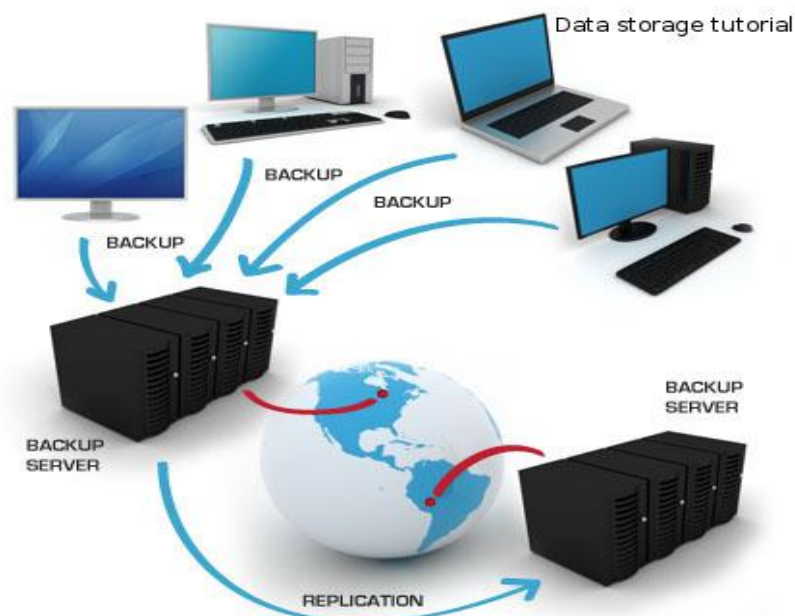
The person who is competent in this CA shall be able to utilize information technology application in accordance with recommended practices.

PENERANGAN/INFORMATION:**1. Backup and Restore**

1.1 It is a fair assumption that everyone has lost a file they needed. Often it is not lost to malware, but either hardware failure or our own failure to backup files. There are tools that will assist you in making backups, or you can manually copy your data to another drive. The advantage to manual backup is that it is fast. The disadvantage is that it is not scheduled. It is not automatic. Whatever the cause of the loss may be, it represents time and effort. Make a backup plan and stick to it.

1.2 There are online services for backing up your files. These services run a program that is scheduled to automatically backup your files (while you sleep). They have an additional advantage of offering off-site backup. A second hard drive in your computer is also vulnerable to malware and an external drive, even when not connected, is susceptible to theft or fire. The biggest disadvantage of such a service is that you are trusting a third party with your data: work, pictures, saved email, etc. Make sure they encrypt your data to stop casual perusal by employees and theft by others. But, remember that they have the encryption key.

1.3 Restore points are a type of backup for the system itself. Create a restore point before installing software, partitioning a drive, etc.



Picture 1.3 Backing up data

2. Authentication

2.1 Authentication refers to the process of making sure a user has access to the resource, usually the entire system; that is, logging on. If you are using accounts, as recommended above, you can also protect your files from other users. They can be given permission to change (delete), read, or denied access. When you attempt to access a resource you do not have permission for, or install a program, you are generally prompted to give permission (and authenticated). You may have to log in as an administrator to accomplish some tasks — then switch back to your user account.

2.2 Standard authentication is accomplished with a password. Pick a password that is hard to guess: more than 8 characters and including non-alphabetical characters, that you can remember. Random would be great, if you have that kind of memory. Otherwise consider modifying a phrase that is meaningful to you (ignore any spaces).

2.3 You also need to manage your password. Change them regularly.

2.4 Don't forget that many passwords are saved on PCs, laptops and smartphones, including banking or online trading passwords. Secure the device first.



Picture 2.4 Authentication in process

3. Biometrics [src]

3.1 Biometrics consists of methods for uniquely recognizing humans based upon one or more intrinsic physical or behavioral traits. It is used as a form of identity access management and access control. Probably the most common form of biometrics use is fingerprint scanning. Other forms of physiological biometrics include, but are not limited to fingerprint, face recognition, DNA, palm print, hand geometry, iris recognition (which has largely replaced retina), and odor/scent. Behavioral biometrics include, but are not limited to typing rhythm, gait, and voice.



Picture 3.2 Types of biometrics

4. Encryption

4.1 Assume your PC (and online backup service) will eventually get hacked. Why not save your files in a way that prevents others from accessing them. Encryption packages are available (some are free) that will create a "container" file that appears like another disk drive to your system. Anything saved on that drive is actually saved in the encrypted container. (You do have to start this up before using it and make sure you close it down when done.) This is a must if you are using business files at home, working with healthcare information, or student grades.

4.2 You can purchase flash drives that will protect your data also. They have the encryption software on them.

5. How to Identify Which Hardware Component is Failing in Your Computer

5.1 Blue Screen 101: Search for the Error Message

- i) This may seem like obvious advice, but searching for information about a blue screen's error message can help immensely. Most blue screens of death you'll encounter on modern versions of Windows will likely be caused by hardware failures. The blue screen of death often displays information about the driver that crashed or the type of error it encountered.
- ii) For example, let's say you encounter a blue screen that identified "NV4_disp.dll" as the driver that caused the blue screen. A quick Google search will reveal that this is the driver for NVIDIA graphics cards, so you now have somewhere to start. It's possible that your graphics card is failing if you encounter such an error message.

5.2 Check Hard Drive SMART Status

- i) Hard drives have a built in S.M.A.R.T. (Self-Monitoring, Analysis, and Reporting Technology) feature. The idea is that the hard drive monitors itself and will notice if it starts to fail, providing you with some advance notice before the drive fails completely. This isn't perfect, so your hard drive may fail even if SMART says everything is okay.
- ii) If you see any sort of "SMART error" message, your hard drive is failing. You can use SMART analysis tools to view the SMART health status information your hard drives are reporting.

5.3 Test Your RAM

- i) RAM failure can result in a variety of problems. If the computer writes data to RAM and the RAM returns different data because it's malfunctioning, you may see application crashes, blue screens, and file system corruption.
- ii) To test your memory and see if it's working properly, use Windows' built-in Memory Diagnostic tool. The Memory Diagnostic tool will write data to every sector of your RAM and read it back afterwards, ensuring that all your RAM is working properly.

5.4 Check Heat Levels

- i) How hot is inside your computer? Overheating can result in blue screens, crashes, and abrupt shut downs. Your computer may be overheating because you're in a very hot location, it's ventilated poorly, a fan has stopped inside your computer, or it's full of dust.
- ii) Your computer monitors its own internal temperatures and you can access this information. It's generally available in your computer's BIOS, but you can also view it with system information utilities such as SpeedFan or Speccy. Check your computer's recommended temperature level and ensure it's within the appropriate range.
- iii) If your computer is overheating, you may see problems only when you're doing something demanding, such as playing a game that stresses your CPU and graphics card. Be sure to keep an eye on how hot your computer gets when it performs these demanding tasks, not only when it's idle.

5.5 Stress Test Your CPU

- i) You can use a utility like Prime95 to stress test your CPU. Such a utility will force your computer's CPU to perform calculations without allowing it to rest, working it hard and generating heat. If your CPU is becoming too hot, you'll start to see errors or system crashes.
- ii) Overclockers use Prime95 to stress test their overclock settings — if Prime95 experiences errors, they throttle back on their overclocks to ensure the CPU runs cooler and more stable. It's a good way to check if your CPU is stable under load.

5.6 Stress Test Your Graphics Card

- i) Your graphics card can also be stress tested. For example, if your graphics driver crashes while playing games, the games themselves crash, or you see odd graphical corruption, you can run a graphics benchmark utility like 3DMark. The benchmark will stress your graphics card and, if it's overheating or failing under load, you'll see graphical problems, crashes, or blue screens while running the benchmark.
- ii) If the benchmark seems to work fine but you have issues playing a certain game, it may just be a problem with that game.

5.7 Swap it Out

- i) Not every hardware problem is easy to diagnose. If you have a bad motherboard or power supply, their problems may only manifest through occasional odd issues with other components. It's hard to tell if these components are causing problems unless you replace them completely.
- ii) Ultimately, the best way to determine whether a component is faulty is to swap it out. For example, if you think your graphics card may be causing your computer to blue screen, pull the graphics card out of your computer and swap in a new graphics card. If everything is working well, it's likely that your previous graphics card was bad.
- iii) This isn't easy for people who don't have boxes of components sitting around, but it's the ideal way to troubleshoot. Troubleshooting is all about trial and error, and swapping components out allows you to pin down which component is actually causing the problem through a process of elimination.

6. Managing Digital Documents

6.1 Digital documents, like paper, need to be stored so they can be retrieved and used as needed. For a paperless office -- or more likely, a near paperless one -- you'll need a way to turn paper documents into digital, or electronic documents.

6.2 That type of communication transformation is handled by a document imaging system. Since not having paper is unlikely, you'll probably find an ongoing need to turn paper into electronic files. A document management system usually goes beyond that transformation to deal with all documents -- computer generated, as well as those that are faxed, scanned or e-mailed.

6.3 The basic document imaging system consists of a scanner and software that allows paper documents to be scanned, converted to electronic images and saved in PDF, TIFF or other formats for storage on CDs, DVDs, a computer hard disk or a network server.

6.4 Most document imaging systems include optical character recognition (OCR). This allows the images to be recognized as text when saved as searchable PDFs or

NO. KOD ICODE NO.	Z-009-4:2015-M08/P(2/4)	MukaSurat / Page : Drpd / of : 11
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copied into programs like Microsoft Word or Excel. With OCR, you also can search for specific words or phrases within a scanned document.

6.5 Document management systems provide the electronic file cabinets and filing system tools. They combine data storage space with software that goes beyond document imaging to include functions such as audit trail and reporting tools, document expiration controls to delete documents at specified dates and Web-based document viewing.

6.6 More advanced document management systems offer extras such as:

- i) User authentication for increased security
- ii) Automatic e-mail notification
- iii) Digital signatures
- iv) Data encryption for secure sending
- v) Custom keys that provide prompts for correct indexing
- vi) Targeted solutions for industries such as health care, financial or manufacturing
- vii) Multimedia filing, bringing together various types of files such as visual images, video, text and spreadsheets.

6.7 If you have large quantities of paper documents to turn into electronic files or need help getting the job done quickly, you may want to outsource document imaging to a company like microMEDIA. One of the company's current on-site conversion projects includes OCR and fully searchable PDF files with more than 25 million images.

6.8 And if you don't have a robust enough server or want to store electronic files off site, many document imaging companies offer a hosted server solution. Web-based document hosting allows you to store or back up your files on their server and then access them securely anywhere, any time. The advantage is that valuable files are off site in case your company's server goes down or cannot be used on site, like during a hurricane.

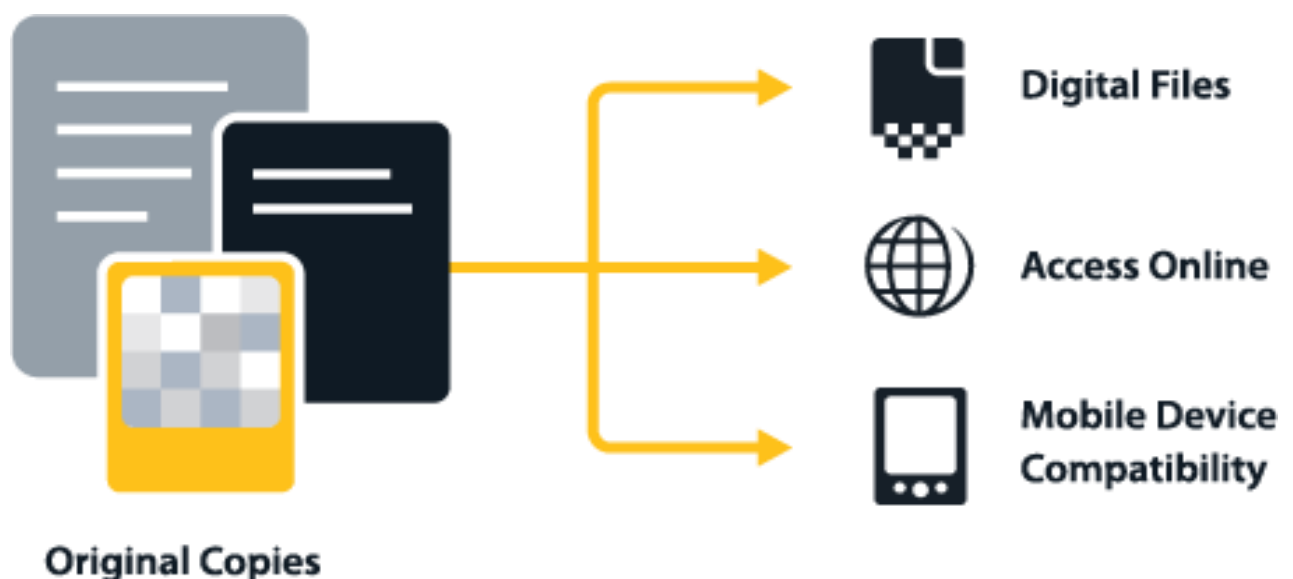
6.7 Beyond your company's paperless needs, employees may have a home office. On the next page, we'll look at how to bring a paperless look to your home office.

NO. KOD ICODE NO.	Z-009-4:2015-M08/P(2/4)	MukaSurat / Page : Drpd / of : 11
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6.8 There are many benefits with a paperless office, beyond simply having less paper to deal with. While you're unlikely to have a completely paper-free office, electronic communication and other systems can eliminate much of it. Let's take a closer look at how this has benefited some businesses:

- i) Reduced costs and quicker access to information. MajesticInvites.com is an eight-person company that designs online invitations. Being paperless saves the company about \$100,000 per year on printing, mailing, paper and storage space, reports Alina Uzilov, the company's president. And because employees can access electronic documents more quickly than paper, they can almost instantly respond to customer questions or make their changes [source: Small Business Computing].
- ii) More space. When he got ready to switch locations after making his office paperless, real estate broker Ed Branson of Branson's California Property noticed that he didn't even have half as many file cabinets as before going paperless [source: Microsoft.com]. Photographer: Peterfactor | Agency: Dreamstime Many doctors' offices are transitioning to electronic medical records.
- iii) Document security and easy information sharing. Obstetrician Rose Kung, M.D., found storing and retrieving paper patient charts was time consuming for her practice at Women's College Hospital in Toronto. Switching to an electronic document management system reduced chart filing and finding time -- and improved security of patient information. Password protection at different levels limits the access that office staff have to the information. Kung can also send electronic copies of patient records to physicians anywhere in the world [source: A&L Computer Software Limited].
- iv) Ability to handle company growth. Transervice Logistics in Lake Success, N.Y., helps companies manage their delivery systems. The human resources staff found they had more than 30,000 pieces of paper to track and store in their 1,000 employee records. With that information now stored electronically, the department was able to handle the addition of 500 employees (potentially 15,000 more pieces of paper) without needing additional space.

- v) Access anywhere and electronic prescriptions. South Shore Skin Care Center, near Boston, has switched to an electronic medical records (EMR) system from paper patient charts. With the EMR, doctors can access patient information from anywhere at any time. They can also send electronic prescriptions directly to pharmacies [source: dBusiness News].
- vi) Less paper in. A company plans to accept paper faxes from customers. However, the company uses electronic faxing to intercept those faxes, turn them into electronic documents and keep them from coming into the company as paper.
- vii) Less temporary paper. At dentist KambizMoin's office in Manchester, N.H., not only are patient charts electronic, but also patient arrivals. A patient who arrives for an appointment types his or her name in a waiting room computer and puts a finger on a biometric reader. Their arrival is then posted on the office computers, color coded to show an early, on-time or late arrival [source: Orthodontic Products Online].
- viii) Less paper out. Tax professional Joseph Anthony in Portland, Ore., files clients tax returns electronically, eliminating paper returns, and then turns the return into a PDF document that he gives the client on a rewriteable CD.



Picture 6.8 Digital documentation

NO. KOD <i>ICODE NO.</i>	Z-009-4:2015-M08/P(2/4)	MukaSurat / Page : Drpd / of : 11
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SOALAN / QUESTION(S):

1. Define authentication.

2. What is the function of Encryption?

RUJUKAN / REFERENCE:

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